

Full Stack Developer Technical Assessment (24-Hour Challenge)

Objective:

Build and deploy a production-ready Task Management Application demonstrating strong understanding of backend architecture, authentication, security practices, database handling, frontend integration, and deployment strategies.

Allowed Tech Stack (Choose Any Combination):

- Backend: Node.js (Express) OR Next.js API Routes
- Frontend: Next.js (Preferred) or React
- Database: MongoDB OR PostgreSQL
- Deployment: AWS / Azure / Render / Railway / Vercel / Any Cloud Platform

Core Requirements:

- User Registration & Login
- JWT-based authentication
- Store access token securely in HTTP-only cookies
- Passwords must be hashed (bcrypt or equivalent)
- CRUD APIs for managing tasks (Title, Description, Status, Created Date)
- Users must access only their own tasks (proper authorization required)
- Proper validation and structured error handling

Security & Advanced Expectations:

- Implement secure cookie configuration (HttpOnly, Secure flags where applicable)
- Encrypt sensitive request or response payload fields (e.g., using AES or similar)
- Prevent common vulnerabilities (basic SQL injection prevention, input validation)
- Use proper HTTP status codes
- Environment variables must not be hardcoded

Functional Expectations:

- Pagination in task listing API
- Filter tasks by status
- Search tasks by title
- Protected frontend routes
- Clean UI (basic but functional)

Deployment Requirements:

- Application must be deployed and publicly accessible
- Provide live URL
- Provide GitHub repository link
- Include README with setup instructions and architecture explanation
- Provide sample API request/response documentation

AI Usage Policy:

- You are allowed to use AI tools (ChatGPT, Copilot, AI Agents, etc.)
- However, you must fully understand and explain your implementation during follow-up discussion
- Direct copy-paste or template-based submissions without understanding will be rejected

Evaluation Criteria:

- Code Structure & Clean Architecture – 20%
- Authentication & Security Implementation – 20%
- Database Design & Query Handling – 15%
- API Design & Error Handling – 15%
- Frontend Integration & UX – 10%
- Deployment & DevOps Understanding – 10%
- Documentation & Clarity – 10%

Submission Guidelines:

- Submit within 24 hours from assignment receipt
- Share GitHub repository link and live deployed URL
- Ensure application runs without manual intervention